

ESTIMATING THE SECOND-ORDER PRODUCT MOMENT FROM THE PROCESS OF COUNTS

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Summary: A suggestion by Brillinger for estimating the
second-order product moment from the process of
counts is shown to improve a standard estimate
in terms of mean squared error. This is proven
analytically in the case of a Poisson process,
and using simulation techniques in some other cases.

1. INTRODUCTION

Let $(N_t)_{t \geq 0}$ be an orderly, stationary point process with rate $p_N = \text{Ed}N_t/\text{dt}$,
where $dN_t = N_{t+\text{dt}} - N_t$, and second-order product moment $p_{NN}(u) = \text{Ed}N_{t+u}dN_t/\text{dtdu}$.
The autointensity function of the process is $h(u) = p_{NN}(u)/p_N$, and can be
thought of as the conditional probability of an event at time $t+u$ given
an event at time t . This is often an important (and physically meaningful)

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parameter in applications such as earthquake prediction, hurricane monitoring etc. It is closely related to the familiar covariance density and the spectral density of counts. For a review of these concepts, see e.g. Brillinger (1978). In order to estimate h , one usually proceeds by estimating p_{NN} , and then dividing by N_T/T , the obvious estimate of p_N from observing $(N_t)_{0 \leq t \leq T}$.

It is sometimes the case that one only gets to observe a sample, typically equispaced, of the path N_t . Denote $Y_k = N_k - N_{k-1}$. This process of counts can be treated as a time series, and analyzed with standard methods of non-parametric time series analysis, but in this note we will be interested in estimating $p_{NN}(u)$ from observing $Y_1, \dots, Y_T$. The relation between time series and point process parameters as well as methods of estimation in the sampled context is discussed in Guttorp and Thompson (1983). The simplest estimate of p_{NN} is

$$(1) \quad \hat{p}_{NN}^{(1)}(u) = \frac{1}{T-u} \sum_{i=1}^{T-u} Y_i Y_{i+u}$$

A competitor, suggested (in a slightly different context) by Brillinger (1975), is

$$(2) \quad \hat{p}_{NN}^{(2)}(u) = \frac{1}{T} \sum_{i=1}^{T-u} Y_i Y_{i+u} + \frac{u}{T} \left(\frac{N_T}{T} \right)^2.$$

The reasoning behind (2) is that for large lags u , events near t and near $t+u$ are virtually uncorrelated in many cases. The choice of divisor (T or $T-u$) is a familiar problem in time series analysis, being a trade-off between bias and variance. Brillinger's discussion suggests that $\hat{p}_{NN}^{(2)}$ may have smaller mean squared error (mse) than $\hat{p}_{NN}^{(1)}$. We will compare these estimates analytically in the Poisson case and using simulation techniques in some non-Poisson cases.

2. THE POISSON CASE

If (N_t) is a Poisson process (by stationarity it must be homogeneous) we

have $p_{NN}(u) = p_N^2$. If we knew that the process were Poisson, we could use the maximum likelihood estimate $(N_T/T)^2$ or the minimum variance unbiased estimate $N_T(N_T-1)/T^2$ (which dominates the mle in terms of mse). The Fisher information about p_N^2 in $(N_t, 0 \leq t \leq T)$ is seen to be $(4p_N^3/T)^{-1}$, and neither of the estimates above are efficient. Straightforward algebra shows that

$$(3) \quad \text{mse}(\hat{p}_{NN}^{(1)}(u)) = \frac{2p_N^3(2T-3u) + p_N^2(T-u)}{(T-u)^2} \quad \text{if } u < T/2$$

$$= \frac{2p_N^3 + p_N^2}{T-u} \quad \text{if } u \geq T/2$$

$$(4) \quad \text{mse}(\hat{p}_{NN}^{(2)}(u)) = \{2p_N^3(2T^4 + uT^3 - 2u^2T^2) + p_N^2(T^4 - uT^3 + 8uT^2 - u^2T) + p_N u^2\}T^{-5} \quad \text{if } u < T/2$$

$$= \{2p_N^3(T^4 + 3uT^3 - 2u^2T^2) + p_N^2(T^4 - uT^3 + 8uT^2 - u^2T) + p_N u^2\}T^{-5} \quad \text{if } u \geq T/2.$$

In the case $u < T/2$, formula (3) was given by Zucchini (1975). For large T and small u the efficiency of each of the $\hat{p}_{NN}^{(\cdot)}$ is approximately $4p_N/(1+4p_N)$. A direct comparison of the exact mse's in (3) and (4) shows that

$$\text{mse}(\hat{p}_{NN}^{(1)}(u)) - \text{mse}(\hat{p}_{NN}^{(2)}(u)) = \frac{2p_N^3\gamma^2T}{(T-u)^2}(2-5\gamma+2\gamma^2) + \frac{p_N^2}{T-u}\gamma(2-\gamma)$$

$$+ O(T^{-1}(T-u)^{-1}), \quad u < T/2$$

$$= \frac{p_N^3}{T-u}(-2\gamma(2-5\gamma+2\gamma^2)) + \frac{p_N^2}{T-u}\gamma(2-\gamma)$$

$$+ O(T^{-1}(T-u)^{-1}), \quad u \geq T/2,$$

where $\gamma = u/T$. From the above it can be seen, in the Poisson case, that

$\hat{p}_{NN}^{(2)}$ improves on $\hat{p}_{NN}^{(1)}$ uniformly in u in terms of mse.

Remark: We have taken the sampling window to have unit length. In the Poisson case using a different sampling window δ is equivalent to changing the rate from p_N to δp_N .

3. SOME NON-POISSON CASES

When the underlying point process is not Poisson, the computations corresponding to formulae (3) and (4) above become much more complicated, and we have been unable to obtain any neat expressions for the difference in mse's. Instead, we have carried out Monte Carlo simulations of some non-Poisson processes where the second-order product moment is easily computable. Perhaps the simplest generalization is to a renewal process, where p_{NN} is proportional to the renewal density. In the case of a gamma renewal process with shape parameter 2 or 3, this can easily be computed from formula (3.8) in Cox and Isham (1980), and is seen to be

$$p_{NN}(u) = (1 - e^{-2u})/4 \quad \text{for } \alpha = 2$$

$$= \frac{1}{9}(1 - (\cos(3^{1/2}u/2) + 3^{1/2}\sin(3^{1/2}u/2))e^{-3u/2}) \quad \text{for } \alpha = 3$$

where α denotes the shape parameter, and the scale parameter is taken to be 1. Tables 1 and 2 show the results of simulating gamma renewal processes with these parameters for 100 time units with 1000 replications. In terms of mse, $\hat{p}_{NN}^{(2)}$ dominates $\hat{p}_{NN}^{(1)}$ over the whole range of values studied.

TABLE 1

Gamma renewal process, $\alpha = 2$

Lag	1	2	3	4	5	25	50	75
$\text{mse}(\hat{p}_{NN}^{(1)})$	0,0037	0,0037	0,0038	0,0040	0,0038	0,0044	0,0052	0,0107
$\text{mse}(\hat{p}_{NN}^{(2)})$	0,0037	0,0036	0,0037	0,0039	0,0037	0,0036	0,0032	0,0031

TABLE 2

Gamma renewal process, $\alpha = 3$

Lag	1	2	3	4	5	25	50	75
$\text{mse}(\hat{p}_{NN}^{(1)})$	0,0007	0,0013	0,0014	0,0013	0,0014	0,0016	0,0021	0,0035
$\text{mse}(\hat{p}_{NN}^{(2)})$	0,0007	0,0012	0,0014	0,0013	0,0013	0,0012	0,0011	0,0009

A generalization of the Poisson process in perhaps a more interesting direction is the exponential moving average process (EMA), where successive intervals are dependent, albeit marginally exponentially distributed. A description of this kind of model can be found in Cox and Isham (1980, p. 62 ff.). Given a sequence ε_i of independent exponential random variables with mean $1/\theta$, and an independent sequence B_i of independent Bernoulli variables with success probability p , the n -th first-order EMA interval is given by

$$Y_n = (1-p)\varepsilon_n + B_n\varepsilon_{n-1}.$$

From formula (3.28) in Cox and Isham (1980) we learn that

$$h(u) = \theta + \frac{p(1-p)\theta}{p-(1-p)^2} \left\{ e^{-\theta u} - e^{\frac{-\theta u}{2-3p+p^2}} \right\} \text{ if } 3p-p^2 \neq 1$$

$$= \theta + (1-2p)\theta^2 u e^{-\theta u} \text{ if } 3p-p^2 = 1.$$

The second-order product moment is obtained by multiplying by the rate.

The results for different values of θ and p are shown in Tables 3 through 6 below. Again, the time interval was 100, and we did 1 000 iterations of the simulation procedure. The results are entirely con-

TABLE 3
EMA(1)-process, $\theta = 1$, $p = 0,9$

Lag	1	2	3	4	5	25	50	75
$\text{mse}(\hat{p}_{NN}^{(1)})$	0,0633	0,0616	0,0596	0,0616	0,0611	0,0742	0,0732	0,1336
$\text{mse}(\hat{p}_{NN}^{(2)})$	0,0630	0,0612	0,0589	0,0607	0,0600	0,0639	0,0557	0,0538

TABLE 4
EMA(1)-process, $\theta = 1$, $p = 0,5$

Lag	1	2	3	4	5	25	50	75
$\text{mse}(\hat{p}_{NN}^{(1)})$	0,0761	0,0711	0,0676	0,0681	0,0674	0,0765	0,0797	0,1386
$\text{mse}(\hat{p}_{NN}^{(2)})$	0,0759	0,0706	0,0669	0,0672	0,0662	0,0674	0,0623	0,0605

TABLE 5

EMA(1)-process, $\theta = 1$, $p = 0,1$

Lag	1	2	3	4	5	25	50	75
$\text{mse}(\hat{p}_{NN}^{(1)})$	0,0584	0,0522	0,0555	0,0562	0,0565	0,0683	0,0649	0,1299
$\text{mse}(\hat{p}_{NN}^{(2)})$	0,0582	0,0519	0,0549	0,0553	0,0554	0,0592	0,0505	0,0510

TABLE 6

EMA(1)-process, $\theta = 0,5$, $p = 0,5$

Lag	1	2	3	4	5	25	50	75
$\text{mse}(\hat{p}_{NN}^{(1)})$	0,5647	0,5377	0,5477	0,5499	0,5478	0,6434	0,5844	1,0904
$\text{mse}(\hat{p}_{NN}^{(2)})$	0,5632	0,5354	0,5439	0,5446	0,5410	0,5785	0,5070	0,5002

sistent with our other findings, in that $\hat{p}_{NN}^{(2)}$ is vastly superior to $\hat{p}_{NN}^{(1)}$ for large lags u , and marginally so for small lags.

Finally, in order to see what happens to the two estimators in a case where we do not have approximate independence for large lags, we simulated a doubly stochastic Poisson process with a rate function that was a realization of the square of a stationary Ornstein-Uhlenbeck process (see Cox and Isham, 1980 and Karlin and Taylor, 1981 for descriptions of these processes). By choosing a small drift parameter for the Ornstein-Uhlenbeck process, the resulting point process is seen to have a slowly varying $p_{NN}(u)$. In fact,

$$p_{NN}(u) = \frac{\sigma^4}{4\beta^2}(2e^{-2\beta u} + 1).$$

Here σ^2 is the infinitesimal variance of the Ornstein-Uhlenbeck process, and β is the drift parameter. To be consistent with previous simulations we used the time interval 100 and 1000 iterations. Because of the high variability and long term dependence of this doubly stochastic Poisson process, one would not expect to be able to estimate $p_{NN}(u)$ very well, without having a quite long realization. This is indeed borne out in the simulations, where it is seen that $\hat{p}_{NN}^{(2)}$ has smaller mean squared error than $\hat{p}_{NN}^{(1)}$ for small and moderate lags. Only for lag 50 is the Cox estimator superior, and then only by a small amount. For larger lags the dependence is again small.

TABLE 7

Doubly stochastic Poisson process, $\beta = 0,05$, $\sigma = 0,3$

Lag	1	2	3	4	5	25	50	75
$mse(\hat{p}_{NN}^{(1)})$	9,67	9,34	8,79	8,50	8,05	2,65	1,40	2,23
$mse(\hat{p}_{NN}^{(2)})$	9,57	9,16	8,54	8,18	7,69	2,58	1,84	2,18

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